

Another part of the data preprocessing was the harmonization of pharmaceutical commodity codes over time. Statistics Canada updates the HS code system every few years. A single product may appear under three to five different codes across the full study period. Without a crosswalk, a simple plot by code shows sharp breaks at each revision year. To handle this, we built a three-stage method that joins each retired code to its current active code. The result is a look-up table that enables the application to display the information for multiple years for a current commodity that had different HS code classifications in previous years.

At the first stage, we link a retired code to a new one using a text-matching script. The new code must start within 12 months of the old one. Their descriptions must also closely match. For example, 1988 heparin trade HS code “3001.90.10.00” moved to HS code “3001.90.00.10” in 1998. The descriptions were identical, so the two were linked. Harder cases proceeded to the second stage, where manual intervention was needed. For instance, blood plasma trade shifted from 3002.10.00.21 to 3002.12.00.12 in 2017. The new description was reworded from “Blood plasma, for therapeutic or prophylactic use in humans” to “Human blood plasma”, which was confirmed manually. The last stage handles what remains using 6-digit HS code transition tables and Canadian sub-code conventions. An example is the 2022 rename of vaccines for veterinary medicine from HS code “3002.30” to “3002.42”. This was applied as a direct override from the World Trade Organization (2026) concordance table. The output gives each historical code a stable family identifier, allowing continuous time series across all HS revisions.

Two structural issues were also identified and addressed. First, the same 10-digit HS code was sometimes reused for unrelated products. For example, code “3002.90.00.20” carried trade in butter-making ferment cultures from 1998 to 2016. The same code was then repurposed in 2022 for toxoids and related biological agents. A simple merge by code would mix these into one series. Our approach detects such reuse and addresses them accordingly. Second, some records lose their code label without a formal termination tag. The time series then looks active when the trade of such commodity has in fact stopped. We address this by inferring termination from the last observed activity when a different variant under the same code begins later. Together, these steps preserve continuity where it should exist and prevent false continuity where it should not.

In total, the crosswalk consolidates 475 distinct import HS commodity codes into 118 families, and 78 export HS commodity codes into 48 families. Also, five import commodity codes and two export commodity codes ended without a 2-digit pharmaceutical HS code successor. This reflects an exit from the pharmaceutical classification to other

classifications. An example is the 2022 move of “diagnostic reagents” to a non-pharmaceutical commodity type, namely miscellaneous chemical products.

Reference:

World Trade Organization. (2026). *WTO Tariff and Trade Data—Official Global Trade Database*. <https://ttd.wto.org/en/resources/reference-nomenclatures>